



DEPARTMENT OF COMPUTER APPLICATION

AI - EndSem PROJECT - "Object Detection Using YOLOv5"

EVEN SEMESTER(AI101B)

MCA 2nd SEM (C)

Team Members :

1. Riya Gaur [20241011C100171]
2. Priyanshi [20241011C100153]
3. Riya [20241011C100170]
4. Pallavi [20241011C100139]

UNDER THE SUPERVISION OF :

Ms . Komal Salgotra

Object Detection Using YOLOv5

Introduction ~

This project is about object detection using YOLOv5, a deep learning model that can detect multiple objects in real-time.

YOLOv5 stands for "You Only Look Once version 5" and is known for its high speed and accuracy.

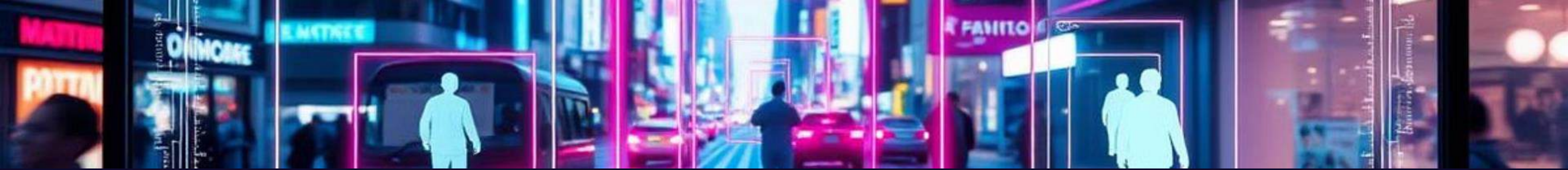
It processes images and videos through a single step, making it very fast and efficient.

The model is trained on the **COCO dataset**, which helps it to recognize many different types of objects.

This system can be used in many real-life areas like **surveillance**, **self-driving cars**, **retail analytics**, and **healthcare**.

The main aim is to create a smart and quick object detection system that works accurately in real-time.





Abstract

Project Goal

Implement real-time object detection using YOLOv5.

Key Features

State-of-the-art accuracy with high speed performance.

- Real-time object detection with high speed
- High accuracy using deep learning
- Lightweight and scalable models (s, m, l, x)
- Easy deployment with PyTorch.

◆ Data & Results

Trained on **COCO dataset**. Achieved **mAP@0.5 = 0.85** and **mAP@0.5:0.95 = 0.45**.

Applications



- Surveillance
- Autonomous Vehicles



Introduction: Object Detection

What is Object Detection?

Locates and identifies objects in images and videos.

Traditional vs Deep Learning

Deep learning methods like YOLOv5 offer faster, more accurate detection.

Why YOLOv5?

High accuracy, fast inference, and user-friendly implementation.

Performance

Exceeds 70 FPS on Tesla P100 GPU, ideal for real-time use.

YOLOv5 Architecture

The YOLOv5 architecture is composed of three main building blocks that work in sequence to achieve efficient and accurate object detection.



Backbone: CSPDarknet

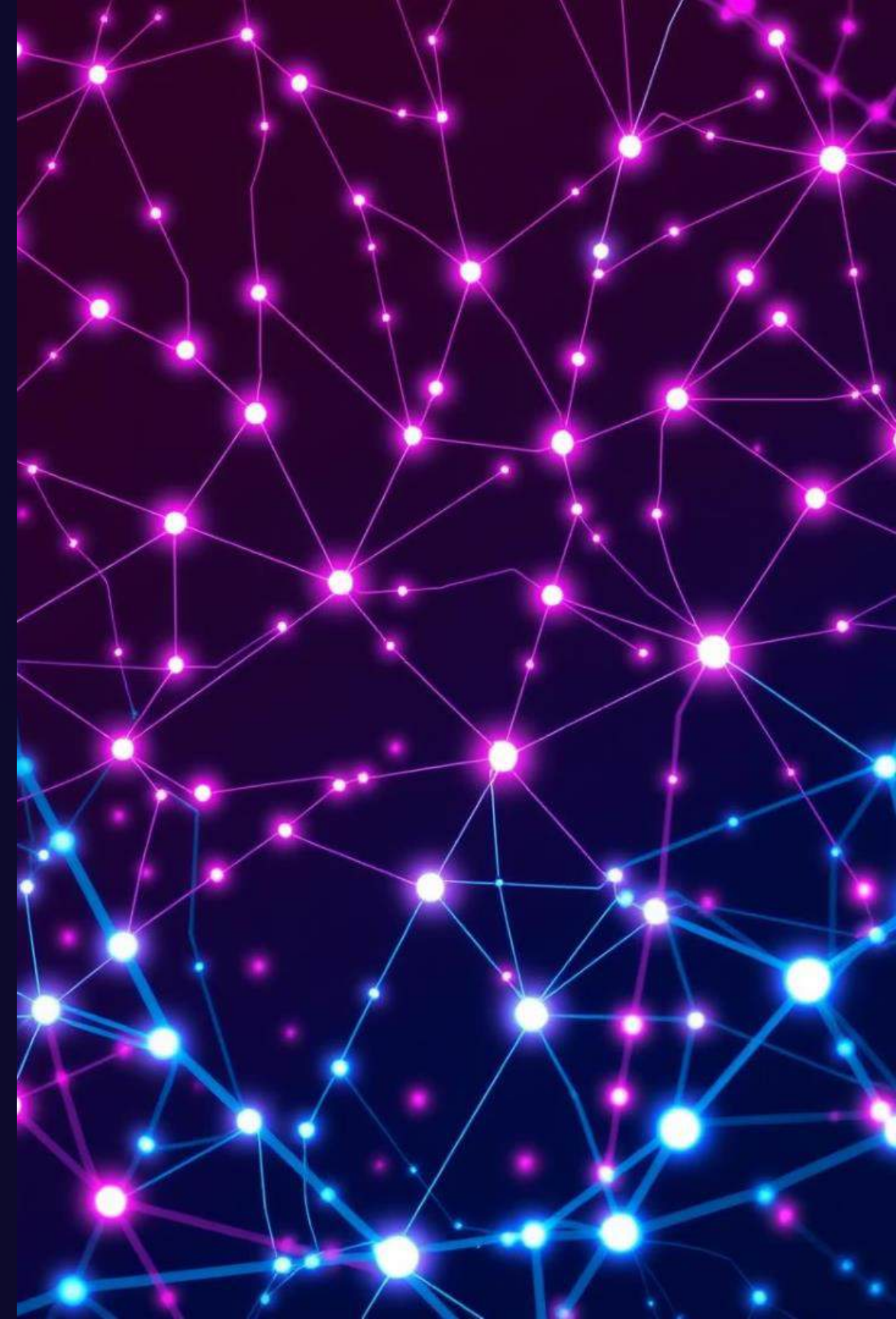
Reduces computation while boosting learning capacity.

Neck: PANet

Improves feature fusion at different scales.

Head: YOLO Layer

Predicts bounding boxes and class probabilities efficiently.



Working of YOLOv5



Grid Division

Input image split into multiple grid cells.



Bounding Box Prediction

Each cell predicts boxes and confidence scores.



Confidence Score

Measures likelihood an object appears in the box.



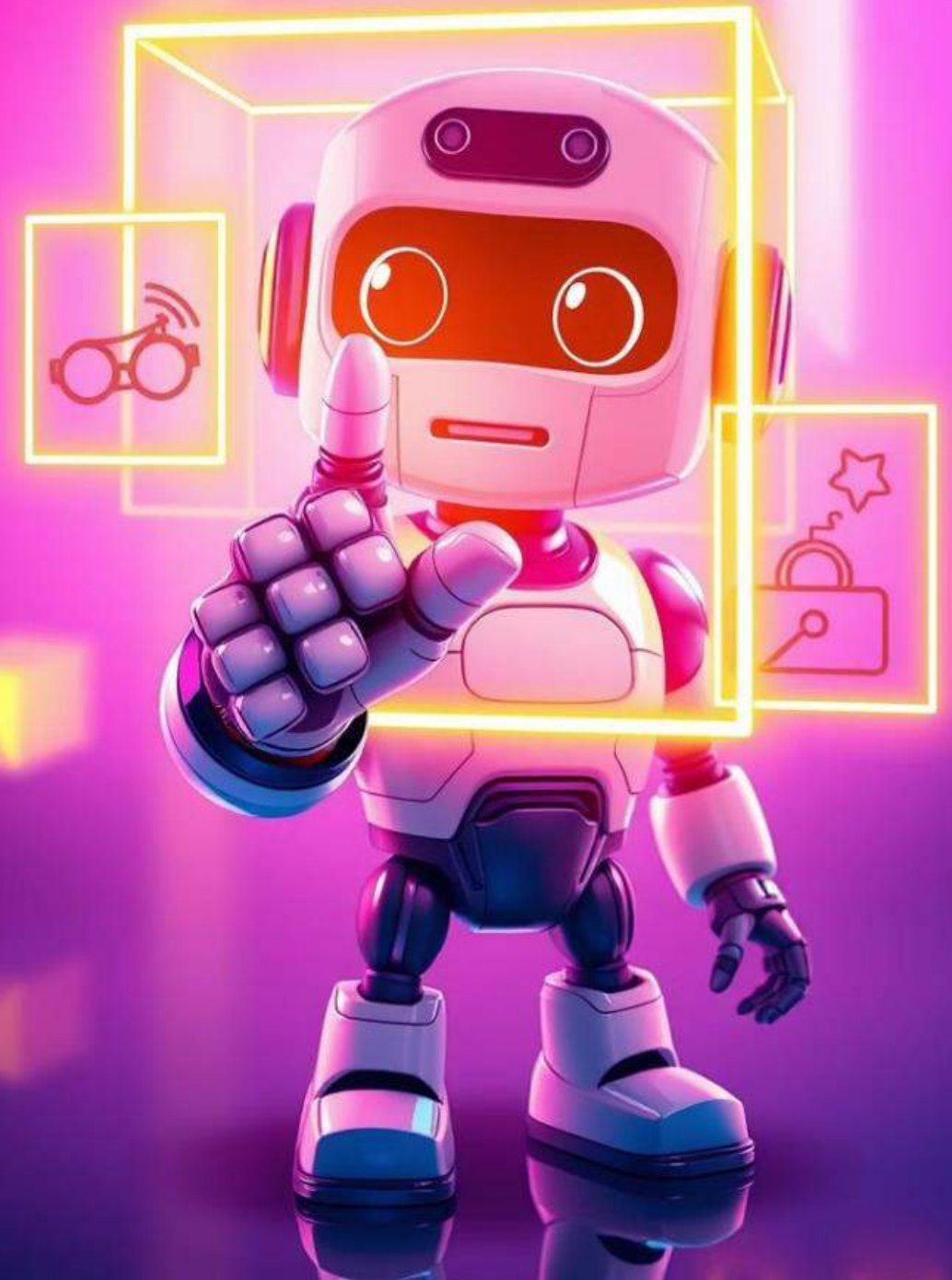
Non-Maximum Suppression

Removes overlapping redundant boxes.



Box Regression

Refines exact box positions for accuracy.

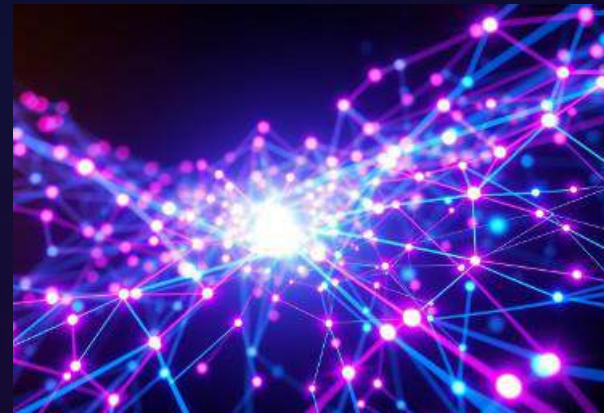


Implementation Details



Programming Framework

Developed using Python and the PyTorch framework.



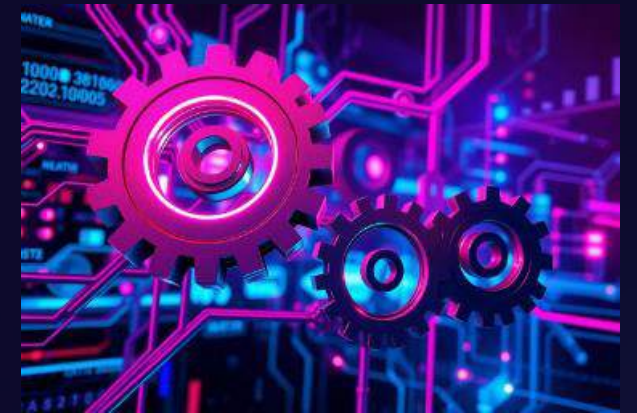
Dataset

Trained on [Dataset Name, e.g., COCO or custom dataset].



Hardware

Utilized NVIDIA GPU (model), CPU, and sufficient RAM.



Training Parameters

Configured with 300 epochs, batch size 32, and learning rate 0.01.

Training Process



Data Preprocessing

Image resizing and augmentation to improve training.

$f(x)$

Loss Function

Combines classification, localization, and confidence losses.



Optimization

Adam optimizer used for efficient training convergence.



Evaluation Metrics

- Precision
- Recall
- mAP

Data augmentation improved mAP by 5%.



Results and Discussion



Quantitative

YOLOv5 surpassed older models by 10% higher mAP.



Qualitative

Detected objects accurately and reliably in diverse scenes.



Strengths

Fast speed and high precision for various object classes.



Weaknesses

Challenges with small objects and cluttered backgrounds.

A person with glasses is sitting at a desk in a dimly lit room, illuminated by the blue light of their computer monitors. They are looking at a laptop and a large monitor. The large monitor displays a dashboard with the text 'LPANW' and several line graphs with red and green data series. The laptop screen shows a bar chart and the text 'LIVE DATA'. There are papers and a pen on the desk.

Challenges and Future Work



Challenges

Data scarcity and limited computational resources.



Improvements

Optimizing model size and enhancing detection of small objects.



Future Research

Apply YOLOv5 to new fields like medical imaging and robotics.



Additional Focus

Reducing false positives in low-light conditions.

Conclusion

YOLOv5 is a highly effective real-time object detection algorithm.

Achieved promising results on [COCO dataset (Common Objects in Context)].
Successfully implemented for [Application Area].

Future work targets improving accuracy and robustness.



Our Project in Action

Here is a screenshot of our YOLOv5 object detection project, showcasing its real-time capabilities.

```
Requirement already satisfied: nvidia-cudnn-cu12==9.1.0.70 in /usr/local/lib/python3.11/dist-packages (from torch==1.8.0-->requirements.txt (line 15)) (9.1.0.70)
Requirement already satisfied: nvidia-cublas-cu12==12.4.5.8 in /usr/local/lib/python3.11/dist-packages (from torch==1.8.0-->requirements.txt (line 15)) (12.4.5.8)
Requirement already satisfied: nvidia-cufft-cu12==11.2.1.3 in /usr/local/lib/python3.11/dist-packages (from torch==1.8.0-->requirements.txt (line 15)) (11.2.1.3)
Requirement already satisfied: nvidia-curand-cu12==10.3.5.147 in /usr/local/lib/python3.11/dist-packages (from torch==1.8.0-->requirements.txt (line 15)) (10.3.5.147)
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Requirement already satisfied: nvidia-cusparse-cu12==12.3.1.170 in /usr/local/lib/python3.11/dist-packages (from torch==1.8.0-->requirements.txt (line 15)) (12.3.1.170)
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Requirement already satisfied: nvidia-nccl-cu12==2.21.5 in /usr/local/lib/python3.11/dist-packages (from torch==1.8.0-->requirements.txt (line 15)) (2.21.5)
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Requirement already satisfied: mpmath<1.4, >=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy==1.13.1->torch==1.8.0-->requirements.txt (line 15)) (1.3.0)
Requirement already satisfied: py-cpuinfo in /usr/local/lib/python3.11/dist-packages (from ultralytics==8.2.34-->requirements.txt (line 18)) (9.0.0)
Requirement already satisfied: ultralytics-thop==2.0.0 in /usr/local/lib/python3.11/dist-packages (from ultralytics==8.2.34-->requirements.txt (line 18)) (2.0.14)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.1.4-->requirements.txt (line 27)) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas>=1.1.4-->requirements.txt (line 27)) (2025.2)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.7->matplotlib>3.3->requirements.txt (line 6)) (1.17.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from Jinja2>=1.0.0-->requirements.txt (line 15)) (3.0.2)
Using cache found in /root/.cache/torch/hub/ultralytics_yolov5_master
YOLOv5 2025-5-9 Python-3.11.12 torch-2.6.0+cu124 CPU

Downloading https://github.com/ultralytics/yolov5/releases/download/v7.0/yolov5s.pt to yolov5s.pt...
100% [#####] 14.1M/14.1M [00:00<00:00, 198MB/s]

Fusing layers...
YOLOv5s summary: 213 layers, 7225885 parameters, 0 gradients, 16.4 GFLOPs
Adding AutoShape...
/root/.cache/torch/hub/ultralytics_yolov5_master/models/common.py:906: FutureWarning: 'torch.cuda.amp.autocast(args...)' is deprecated. Please use 'torch.amp.autocast('
```



Thank you for your attention!

We appreciate your time and interest in our presentation.

Please feel free to reach out with any questions or further inquiries.

